

# Homework 8

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## Part 1: Stepwise Regression

```
rm(list = ls())  
uscrime = read.table('uscrime.txt', header = TRUE)
```

**Backward Elimination.** Use trace argument to suppress the output.

```
model_back = lm(Crime~., data = uscrime)  
step(model_back,  
      direction = 'backward')
```

```
## Start:  AIC=514.65  
## Crime ~ M + So + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 +  
##      U2 + Wealth + Ineq + Prob + Time  
##  
##           Df Sum of Sq      RSS      AIC  
## - So       1         29 1354974 512.65  
## - LF       1        8917 1363862 512.96  
## - Time     1       10304 1365250 513.00  
## - Pop      1       14122 1369068 513.14  
## - NW       1       18395 1373341 513.28  
## - M.F      1       31967 1386913 513.74  
## - Wealth   1       37613 1392558 513.94  
## - Po2      1       37919 1392865 513.95  
## <none>                1354946 514.65  
## - U1       1       83722 1438668 515.47  
## - Po1      1      144306 1499252 517.41  
## - U2       1      181536 1536482 518.56  
## - M        1      193770 1548716 518.93  
## - Prob     1      199538 1554484 519.11  
## - Ed       1      402117 1757063 524.86  
## - Ineq     1      423031 1777977 525.42  
##  
## Step:  AIC=512.65  
## Crime ~ M + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 + U2 +  
##      Wealth + Ineq + Prob + Time  
##
```

```

##          Df Sum of Sq      RSS      AIC
## - Time    1    10341 1365315 511.01
## - LF       1    10878 1365852 511.03
## - Pop      1    14127 1369101 511.14
## - NW       1    21626 1376600 511.39
## - M.F      1    32449 1387423 511.76
## - Po2      1    37954 1392929 511.95
## - Wealth   1    39223 1394197 511.99
## <none>                1354974 512.65
## - U1       1    96420 1451395 513.88
## - Po1      1   144302 1499277 515.41
## - U2       1   189859 1544834 516.81
## - M        1   195084 1550059 516.97
## - Prob     1   204463 1559437 517.26
## - Ed       1   403140 1758114 522.89
## - Ineq     1   488834 1843808 525.13
##
## Step:  AIC=511.01
## Crime ~ M + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 + U2 +
##      Wealth + Ineq + Prob
##
##          Df Sum of Sq      RSS      AIC
## - LF       1    10533 1375848 509.37
## - NW       1    15482 1380797 509.54
## - Pop      1    21846 1387161 509.75
## - Po2      1    28932 1394247 509.99
## - Wealth   1    36070 1401385 510.23
## - M.F      1    41784 1407099 510.42
## <none>                1365315 511.01
## - U1       1    91420 1456735 512.05
## - Po1      1   134137 1499452 513.41
## - U2       1   184143 1549458 514.95
## - M        1   186110 1551425 515.01
## - Prob     1   237493 1602808 516.54
## - Ed       1   409448 1774763 521.33
## - Ineq     1   502909 1868224 523.75
##
## Step:  AIC=509.37
## Crime ~ M + Ed + Po1 + Po2 + M.F + Pop + NW + U1 + U2 + Wealth +
##      Ineq + Prob
##
##          Df Sum of Sq      RSS      AIC
## - NW       1    11675 1387523 507.77
## - Po2      1    21418 1397266 508.09
## - Pop      1    27803 1403651 508.31
## - M.F      1    31252 1407100 508.42
## - Wealth   1    35035 1410883 508.55
## <none>                1375848 509.37
## - U1       1    80954 1456802 510.06
## - Po1      1   123896 1499744 511.42
## - U2       1   190746 1566594 513.47
## - M        1   217716 1593564 514.27
## - Prob     1   226971 1602819 514.54
## - Ed       1   413254 1789103 519.71

```

```

## - Ineq    1    500944 1876792 521.96
##
## Step: AIC=507.77
## Crime ~ M + Ed + Po1 + Po2 + M.F + Pop + U1 + U2 + Wealth + Ineq +
## Prob
##
##      Df Sum of Sq    RSS    AIC
## - Po2    1    16706 1404229 506.33
## - Pop    1    25793 1413315 506.63
## - M.F    1    26785 1414308 506.66
## - Wealth  1    31551 1419073 506.82
## <none>                1387523 507.77
## - U1     1    83881 1471404 508.52
## - Po1    1   118348 1505871 509.61
## - U2     1   201453 1588976 512.14
## - Prob   1   216760 1604282 512.59
## - M      1   309214 1696737 515.22
## - Ed     1   402754 1790276 517.74
## - Ineq   1   589736 1977259 522.41
##
## Step: AIC=506.33
## Crime ~ M + Ed + Po1 + M.F + Pop + U1 + U2 + Wealth + Ineq +
## Prob
##
##      Df Sum of Sq    RSS    AIC
## - Pop    1    22345 1426575 505.07
## - Wealth  1    32142 1436371 505.39
## - M.F    1    36808 1441037 505.54
## <none>                1404229 506.33
## - U1     1    86373 1490602 507.13
## - U2     1   205814 1610043 510.76
## - Prob   1   218607 1622836 511.13
## - M      1   307001 1711230 513.62
## - Ed     1   389502 1793731 515.83
## - Ineq   1   608627 2012856 521.25
## - Po1    1  1050202 2454432 530.57
##
## Step: AIC=505.07
## Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Wealth + Ineq + Prob
##
##      Df Sum of Sq    RSS    AIC
## - Wealth  1    26493 1453068 503.93
## <none>                1426575 505.07
## - M.F    1    84491 1511065 505.77
## - U1     1    99463 1526037 506.24
## - Prob   1   198571 1625145 509.20
## - U2     1   208880 1635455 509.49
## - M      1   320926 1747501 512.61
## - Ed     1   386773 1813348 514.35
## - Ineq   1   594779 2021354 519.45
## - Po1    1  1127277 2553852 530.44
##
## Step: AIC=503.93
## Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob

```

```
##
##           Df Sum of Sq      RSS      AIC
## <none>                1453068 503.93
## - M.F      1      103159 1556227 505.16
## - U1       1      127044 1580112 505.87
## - Prob    1      247978 1701046 509.34
## - U2      1      255443 1708511 509.55
## - M       1      296790 1749858 510.67
## - Ed      1      445788 1898855 514.51
## - Ineq    1      738244 2191312 521.24
## - Po1     1     1672038 3125105 537.93

##
## Call:
## lm(formula = Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob,
##     data = uscrime)
##
## Coefficients:
## (Intercept)          M          Ed          Po1          M.F          U1
##    -6426.10      93.32     180.12     102.65      22.34    -6086.63
##          U2      Ineq      Prob
##      187.35      61.33    -3796.03
```

```
step(model_back,
      direction = 'backward',
      trace = 0)
```

```
##
## Call:
## lm(formula = Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob,
##     data = uscrime)
##
## Coefficients:
## (Intercept)          M          Ed          Po1          M.F          U1
##    -6426.10      93.32     180.12     102.65      22.34    -6086.63
##          U2      Ineq      Prob
##      187.35      61.33    -3796.03
```

In the code above, I performed backward selection. Backward starts with the full model with all of the variables and removes the variable one at a time to reduce the AIC. As a result, our final model consists of the variables: M, Ed, Po1, M.F, U1, U2, Ineq, and Prob. The AIC is 503.93, which is the lowest out of all of them.

## Forward Selection

```
model_forward = lm(Crime~1, data = uscrime)
step(model_forward,
      scope = formula(lm(Crime~., data = uscrime)),
      direction = 'forward',
      k = 2)
```

```

## Start:  AIC=561.02
## Crime ~ 1
##
##      Df Sum of Sq    RSS    AIC
## + Po1      1   3253302 3627626 532.94
## + Po2      1   3058626 3822302 535.39
## + Wealth  1   1340152 5540775 552.84
## + Prob    1   1257075 5623853 553.54
## + Pop     1    783660 6097267 557.34
## + Ed      1    717146 6163781 557.85
## + M.F     1    314867 6566061 560.82
## <none>                6880928 561.02
## + LF      1    245446 6635482 561.32
## + Ineq    1    220530 6660397 561.49
## + U2      1    216354 6664573 561.52
## + Time    1    154545 6726383 561.96
## + So      1     56527 6824400 562.64
## + M       1     55084 6825844 562.65
## + U1      1     17533 6863395 562.90
## + NW      1      7312 6873615 562.97
##
## Step:  AIC=532.94
## Crime ~ Po1
##
##      Df Sum of Sq    RSS    AIC
## + Ineq    1    739819 2887807 524.22
## + M       1    616741 3010885 526.18
## + M.F     1    250522 3377104 531.57
## + NW      1    232434 3395192 531.82
## + So      1    219098 3408528 532.01
## + Wealth  1    180872 3446754 532.53
## <none>                3627626 532.94
## + Po2     1    146167 3481459 533.00
## + Prob    1     92278 3535348 533.72
## + LF      1     77479 3550147 533.92
## + Time    1     43185 3584441 534.37
## + U2      1     17848 3609778 534.70
## + Pop     1      5666 3621959 534.86
## + U1      1      2878 3624748 534.90
## + Ed      1       767 3626859 534.93
##
## Step:  AIC=524.22
## Crime ~ Po1 + Ineq
##
##      Df Sum of Sq    RSS    AIC
## + Ed      1    587050 2300757 515.53
## + M.F     1    454545 2433262 518.17
## + Prob    1    280690 2607117 521.41
## + LF      1    260571 2627236 521.77
## + Wealth  1    213937 2673871 522.60
## + M       1    181236 2706571 523.17
## + Pop     1    130377 2757430 524.04
## <none>                2887807 524.22
## + NW      1     36439 2851369 525.62

```

```

## + So      1      33738 2854069 525.66
## + Po2     1      30673 2857134 525.71
## + U1      1       2309 2885498 526.18
## + Time    1       497 2887310 526.21
## + U2      1       253 2887554 526.21
##
## Step:  AIC=515.53
## Crime ~ Po1 + Ineq + Ed
##
##           Df Sum of Sq      RSS      AIC
## + M       1     239405 2061353 512.37
## + Prob    1     234981 2065776 512.47
## + M.F     1     117026 2183731 515.08
## <none>                2300757 515.53
## + Wealth  1       79540 2221218 515.88
## + U2      1       62112 2238646 516.25
## + Time    1       61770 2238987 516.26
## + Po2     1       42584 2258174 516.66
## + Pop     1       39319 2261438 516.72
## + U1      1        7365 2293392 517.38
## + LF      1        7254 2293503 517.39
## + NW      1        4210 2296547 517.45
## + So      1        4135 2296622 517.45
##
## Step:  AIC=512.37
## Crime ~ Po1 + Ineq + Ed + M
##
##           Df Sum of Sq      RSS      AIC
## + Prob    1     258063 1803290 508.08
## + U2      1     200988 1860365 509.55
## + Wealth  1     163378 1897975 510.49
## <none>                2061353 512.37
## + M.F     1       74398 1986955 512.64
## + U1      1       50835 2010518 513.20
## + Po2     1       45392 2015961 513.32
## + Time    1       42746 2018607 513.39
## + NW      1       16488 2044865 513.99
## + Pop     1        8101 2053251 514.19
## + So      1        3189 2058164 514.30
## + LF      1        2988 2058365 514.30
##
## Step:  AIC=508.08
## Crime ~ Po1 + Ineq + Ed + M + Prob
##
##           Df Sum of Sq      RSS      AIC
## + U2      1     192233 1611057 504.79
## + Wealth  1       86490 1716801 507.77
## + M.F     1       84509 1718781 507.83
## <none>                1803290 508.08
## + U1      1       52313 1750977 508.70
## + Pop     1       47719 1755571 508.82
## + Po2     1       37967 1765323 509.08
## + So      1       21971 1781320 509.51
## + Time    1       10194 1793096 509.82

```

```
## + LF      1      990 1802301 510.06
## + NW      1      797 1802493 510.06
##
## Step:  AIC=504.79
## Crime ~ Po1 + Ineq + Ed + M + Prob + U2
##
##           Df Sum of Sq      RSS      AIC
## <none>                1611057 504.79
## + Wealth  1      59910 1551147 505.00
## + U1      1      54830 1556227 505.16
## + Pop     1      51320 1559737 505.26
## + M.F    1      30945 1580112 505.87
## + Po2     1      25017 1586040 506.05
## + So      1      17958 1593098 506.26
## + LF      1      13179 1597878 506.40
## + Time    1       7159 1603898 506.58
## + NW      1       359 1610698 506.78

##
## Call:
## lm(formula = Crime ~ Po1 + Ineq + Ed + M + Prob + U2, data = uscrime)
##
## Coefficients:
## (Intercept)      Po1      Ineq      Ed      M      Prob
## -5040.50      115.02      67.65     196.47    105.02   -3801.84
##           U2
##           89.37
```

In the code above, we performed forward selection. For forward selection, we start with the model with just the intercept and adds variables one at a time to reduce the AIC the most. The final model consists of the variables: Po1, Ineq, Ed, M, Prob, U2. The final model's AIC is 504.79.

## Stepwise Regression

```
model_step = lm(Crime~., data = uscrime)
step(model_step,
      scope = list(lower = formula(lm(Crime ~ 1, data = uscrime)),
                    upper = formula(lm(Crime~., data = uscrime))),
      direction = 'both')
```

```
## Start:  AIC=514.65
## Crime ~ M + So + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 +
##           U2 + Wealth + Ineq + Prob + Time
##
##           Df Sum of Sq      RSS      AIC
## - So      1         29 1354974 512.65
## - LF      1        8917 1363862 512.96
## - Time    1       10304 1365250 513.00
## - Pop     1       14122 1369068 513.14
## - NW      1       18395 1373341 513.28
```

```

## - M.F      1      31967 1386913 513.74
## - Wealth   1      37613 1392558 513.94
## - Po2      1      37919 1392865 513.95
## <none>          1354946 514.65
## - U1       1      83722 1438668 515.47
## - Po1      1     144306 1499252 517.41
## - U2       1     181536 1536482 518.56
## - M        1     193770 1548716 518.93
## - Prob     1     199538 1554484 519.11
## - Ed       1     402117 1757063 524.86
## - Ineq     1     423031 1777977 525.42
##
## Step:  AIC=512.65
## Crime ~ M + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 + U2 +
##      Wealth + Ineq + Prob + Time
##
##      Df Sum of Sq      RSS      AIC
## - Time      1      10341 1365315 511.01
## - LF        1      10878 1365852 511.03
## - Pop       1      14127 1369101 511.14
## - NW        1      21626 1376600 511.39
## - M.F       1      32449 1387423 511.76
## - Po2       1      37954 1392929 511.95
## - Wealth    1      39223 1394197 511.99
## <none>          1354974 512.65
## - U1        1      96420 1451395 513.88
## + So         1         29 1354946 514.65
## - Po1       1     144302 1499277 515.41
## - U2        1     189859 1544834 516.81
## - M         1     195084 1550059 516.97
## - Prob      1     204463 1559437 517.26
## - Ed        1     403140 1758114 522.89
## - Ineq      1     488834 1843808 525.13
##
## Step:  AIC=511.01
## Crime ~ M + Ed + Po1 + Po2 + LF + M.F + Pop + NW + U1 + U2 +
##      Wealth + Ineq + Prob
##
##      Df Sum of Sq      RSS      AIC
## - LF        1      10533 1375848 509.37
## - NW        1      15482 1380797 509.54
## - Pop       1      21846 1387161 509.75
## - Po2       1      28932 1394247 509.99
## - Wealth    1      36070 1401385 510.23
## - M.F       1      41784 1407099 510.42
## <none>          1365315 511.01
## - U1        1      91420 1456735 512.05
## + Time      1      10341 1354974 512.65
## + So         1         65 1365250 513.00
## - Po1       1     134137 1499452 513.41
## - U2        1     184143 1549458 514.95
## - M         1     186110 1551425 515.01
## - Prob      1     237493 1602808 516.54
## - Ed        1     409448 1774763 521.33

```

```

## - Ineq      1      502909 1868224 523.75
##
## Step:  AIC=509.37
## Crime ~ M + Ed + Po1 + Po2 + M.F + Pop + NW + U1 + U2 + Wealth +
##      Ineq + Prob
##
##           Df Sum of Sq      RSS      AIC
## - NW      1      11675 1387523 507.77
## - Po2     1      21418 1397266 508.09
## - Pop     1      27803 1403651 508.31
## - M.F    1      31252 1407100 508.42
## - Wealth  1      35035 1410883 508.55
## <none>                1375848 509.37
## - U1      1      80954 1456802 510.06
## + LF      1      10533 1365315 511.01
## + Time    1       9996 1365852 511.03
## + So      1       3046 1372802 511.26
## - Po1     1     123896 1499744 511.42
## - U2      1     190746 1566594 513.47
## - M       1     217716 1593564 514.27
## - Prob    1     226971 1602819 514.54
## - Ed      1     413254 1789103 519.71
## - Ineq    1     500944 1876792 521.96
##
## Step:  AIC=507.77
## Crime ~ M + Ed + Po1 + Po2 + M.F + Pop + U1 + U2 + Wealth + Ineq +
##      Prob
##
##           Df Sum of Sq      RSS      AIC
## - Po2     1      16706 1404229 506.33
## - Pop     1      25793 1413315 506.63
## - M.F    1      26785 1414308 506.66
## - Wealth  1      31551 1419073 506.82
## <none>                1387523 507.77
## - U1      1      83881 1471404 508.52
## + NW      1      11675 1375848 509.37
## + So      1       7207 1380316 509.52
## + LF      1       6726 1380797 509.54
## + Time    1       4534 1382989 509.61
## - Po1     1     118348 1505871 509.61
## - U2      1     201453 1588976 512.14
## - Prob    1     216760 1604282 512.59
## - M       1     309214 1696737 515.22
## - Ed      1     402754 1790276 517.74
## - Ineq    1     589736 1977259 522.41
##
## Step:  AIC=506.33
## Crime ~ M + Ed + Po1 + M.F + Pop + U1 + U2 + Wealth + Ineq +
##      Prob
##
##           Df Sum of Sq      RSS      AIC
## - Pop     1      22345 1426575 505.07
## - Wealth  1      32142 1436371 505.39
## - M.F    1      36808 1441037 505.54

```

```

## <none>          1404229 506.33
## - U1           1      86373 1490602 507.13
## + Po2          1      16706 1387523 507.77
## + NW           1       6963 1397266 508.09
## + So           1       3807 1400422 508.20
## + LF           1       1986 1402243 508.26
## + Time         1        575 1403654 508.31
## - U2           1     205814 1610043 510.76
## - Prob         1     218607 1622836 511.13
## - M            1     307001 1711230 513.62
## - Ed           1     389502 1793731 515.83
## - Ineq         1     608627 2012856 521.25
## - Po1          1    1050202 2454432 530.57
##
## Step:  AIC=505.07
## Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Wealth + Ineq + Prob
##
##           Df Sum of Sq      RSS      AIC
## - Wealth  1      26493 1453068 503.93
## <none>          1426575 505.07
## - M.F      1     84491 1511065 505.77
## - U1       1     99463 1526037 506.24
## + Pop      1     22345 1404229 506.33
## + Po2      1     13259 1413315 506.63
## + NW       1      5927 1420648 506.87
## + So       1      5724 1420851 506.88
## + LF       1      5176 1421398 506.90
## + Time     1      3913 1422661 506.94
## - Prob     1     198571 1625145 509.20
## - U2       1     208880 1635455 509.49
## - M        1     320926 1747501 512.61
## - Ed       1     386773 1813348 514.35
## - Ineq     1     594779 2021354 519.45
## - Po1      1    1127277 2553852 530.44
##
## Step:  AIC=503.93
## Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob
##
##           Df Sum of Sq      RSS      AIC
## <none>          1453068 503.93
## + Wealth  1      26493 1426575 505.07
## - M.F      1    103159 1556227 505.16
## + Pop      1     16697 1436371 505.39
## + Po2      1     14148 1438919 505.47
## + So       1      9329 1443739 505.63
## + LF       1      4374 1448694 505.79
## + NW       1      3799 1449269 505.81
## + Time     1      2293 1450775 505.86
## - U1       1    127044 1580112 505.87
## - Prob     1    247978 1701046 509.34
## - U2       1    255443 1708511 509.55
## - M        1    296790 1749858 510.67
## - Ed       1    445788 1898855 514.51
## - Ineq     1    738244 2191312 521.24

```

```
## - Po1      1   1672038 3125105 537.93
```

```
##
```

```
## Call:
```

```
## lm(formula = Crime ~ M + Ed + Po1 + M.F + U1 + U2 + Ineq + Prob,  
##      data = uscrime)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)          M          Ed          Po1          M.F          U1  
##   -6426.10       93.32      180.12      102.65       22.34     -6086.63  
##           U2       Ineq       Prob  
##      187.35      61.33     -3796.03
```

```
summary(model_back)
```

```
##
```

```
## Call:
```

```
## lm(formula = Crime ~ ., data = uscrime)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -395.74  -98.09   -6.69   112.99   512.67
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept) -5.984e+03  1.628e+03  -3.675 0.000893 ***  
## M            8.783e+01  4.171e+01   2.106 0.043443 *  
## So          -3.803e+00  1.488e+02  -0.026 0.979765  
## Ed           1.883e+02  6.209e+01   3.033 0.004861 **  
## Po1          1.928e+02  1.061e+02   1.817 0.078892 .  
## Po2         -1.094e+02  1.175e+02  -0.931 0.358830  
## LF          -6.638e+02  1.470e+03  -0.452 0.654654  
## M.F          1.741e+01  2.035e+01   0.855 0.398995  
## Pop         -7.330e-01  1.290e+00  -0.568 0.573845  
## NW           4.204e+00  6.481e+00   0.649 0.521279  
## U1          -5.827e+03  4.210e+03  -1.384 0.176238  
## U2           1.678e+02  8.234e+01   2.038 0.050161 .  
## Wealth       9.617e-02  1.037e-01   0.928 0.360754  
## Ineq         7.067e+01  2.272e+01   3.111 0.003983 **  
## Prob        -4.855e+03  2.272e+03  -2.137 0.040627 *  
## Time        -3.479e+00  7.165e+00  -0.486 0.630708
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 209.1 on 31 degrees of freedom
```

```
## Multiple R-squared:  0.8031, Adjusted R-squared:  0.7078
```

```
## F-statistic: 8.429 on 15 and 31 DF,  p-value: 3.539e-07
```

```
summary(model_step)
```

```
##
```

```
## Call:
```

```
## lm(formula = Crime ~ ., data = uscrime)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -395.74  -98.09   -6.69  112.99  512.67
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.984e+03  1.628e+03  -3.675 0.000893 ***
## M             8.783e+01  4.171e+01   2.106 0.043443 *
## So            -3.803e+00  1.488e+02  -0.026 0.979765
## Ed             1.883e+02  6.209e+01   3.033 0.004861 **
## Po1            1.928e+02  1.061e+02   1.817 0.078892 .
## Po2           -1.094e+02  1.175e+02  -0.931 0.358830
## LF            -6.638e+02  1.470e+03  -0.452 0.654654
## M.F            1.741e+01  2.035e+01   0.855 0.398995
## Pop           -7.330e-01  1.290e+00  -0.568 0.573845
## NW             4.204e+00  6.481e+00   0.649 0.521279
## U1            -5.827e+03  4.210e+03  -1.384 0.176238
## U2             1.678e+02  8.234e+01   2.038 0.050161 .
## Wealth        9.617e-02  1.037e-01   0.928 0.360754
## Ineq           7.067e+01  2.272e+01   3.111 0.003983 **
## Prob          -4.855e+03  2.272e+03  -2.137 0.040627 *
## Time          -3.479e+00  7.165e+00  -0.486 0.630708
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 209.1 on 31 degrees of freedom
## Multiple R-squared:  0.8031, Adjusted R-squared:  0.7078
## F-statistic: 8.429 on 15 and 31 DF, p-value: 3.539e-07
```

In the code above, we perform stepwise elimination (both directions). Stepwise elimination starts with the full model and looks at both additions and removals to the model at each step. The final model consists of the variables: M, Ed, Po1, M.F, U1, U2, Ineq, and Prob. The final model's AIC is 503.93.

## LASSO

First, I scaled the data.

```
x = as.matrix(uscrime[, -which(names(uscrime) == 'Crime')])
y = uscrime$Crime
x_scaled = scale(x)
```

Next, I used `cv.glmnet()` to create my lasso model.

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

```
## Loading required package: Matrix
```

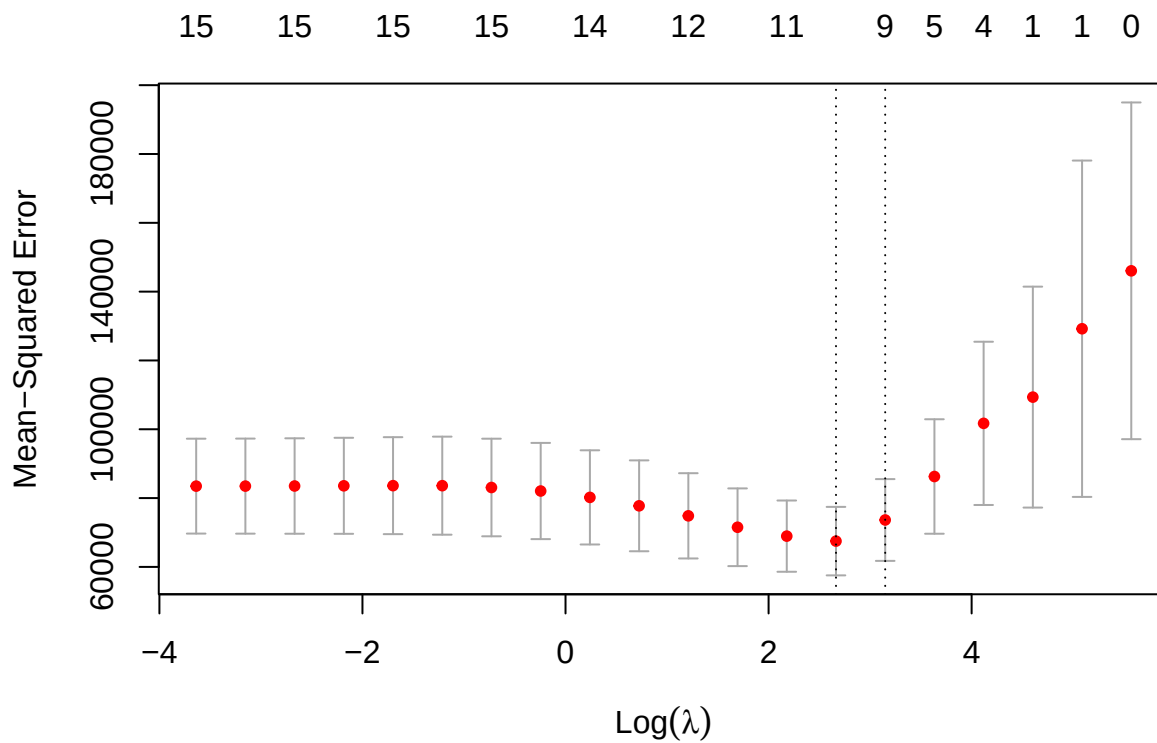
```
## Loaded glmnet 4.1-9
```

```
set.seed(123)
model_lasso = cv.glmnet(x = as.matrix(uscrime[, -16]),
                        y = as.matrix(uscrime[, 16]),
                        alpha = 1,
                        nfolds = 8,
                        nlambda = 20,
                        type.measure = "mse",
                        family = "gaussian",
                        standardize = TRUE)
```

```
model_lasso
```

```
##
## Call:  cv.glmnet(x = as.matrix(uscrime[, -16]), y = as.matrix(uscrime[, 16]), type.measure = "mse")
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min 14.35      7  67493  9950      10
## 1se 23.31      6  73636 11885      9
```

```
plot(model_lasso)
```



```
model_lasso$lambda.min
```

```
## [1] 14.35342
```

```
cbind(model_lasso$lambda, model_lasso$cvm, model_lasso$nzzero)
```

```
##           [,1]      [,2] [,3]
## s0  263.09539664 146048.61    0
## s1  162.02682936 129217.62    1
## s2   99.78393301 109346.07    1
## s3   61.45175663 101715.84    4
## s4   37.84495439  86271.75    5
## s5   23.30674746  73636.30    9
## s6   14.35341873  67492.91   10
## s7    8.83952725  68943.39   11
## s8    5.44380704  71522.60   12
## s9    3.35255883  74841.07   12
## s10   2.06466736  77754.13   13
## s11   1.27152170  80201.74   14
## s12   0.78306436  82052.77   15
## s13   0.48224879  83079.55   15
## s14   0.29699205  83606.83   15
## s15   0.18290202  83614.98   15
## s16   0.11263988  83568.64   14
## s17   0.06936907  83510.97   15
## s18   0.04272082  83476.84   15
## s19   0.02630954  83471.15   15
```

```
coef(model_lasso, s = model_lasso$lambda.min)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##           s0
## (Intercept) -4188.8210567
## M           62.5173281
## So          46.3065171
## Ed          89.8715021
## Po1        105.0118443
## Po2         .
## LF          80.0219314
## M.F        15.7901041
## Pop         .
## NW          0.2279318
## U1          .
## U2         35.8812138
## Wealth     .
## Ineq       41.6921408
## Prob     -3404.8060716
## Time       .
```

As you can see in the plot above, the minimum MSE occurs at  $\log(\lambda)$  when it is approximately equal to 2.2. At this  $\lambda$ , the model keeps 10 variables.

The right dashed line's  $\lambda$  is slightly larger. It also has a slightly larger MSE, and it uses 9 variables.

The output with the coefficients show the non-zero variables (coefficients) of the  $\lambda_{\min}$ . The coefficients are M, So, Ed, Po1, LF, M.F, NW, U2, Ineq, Prob. I will be focusing on prediction performance, so I will focus on using  $\lambda_{\min}$ . These are the 10 variables.

```
min(model_lasso$cvm)
```

```
## [1] 67492.91
```

The RMSE of the lambda minimum is 193.16.

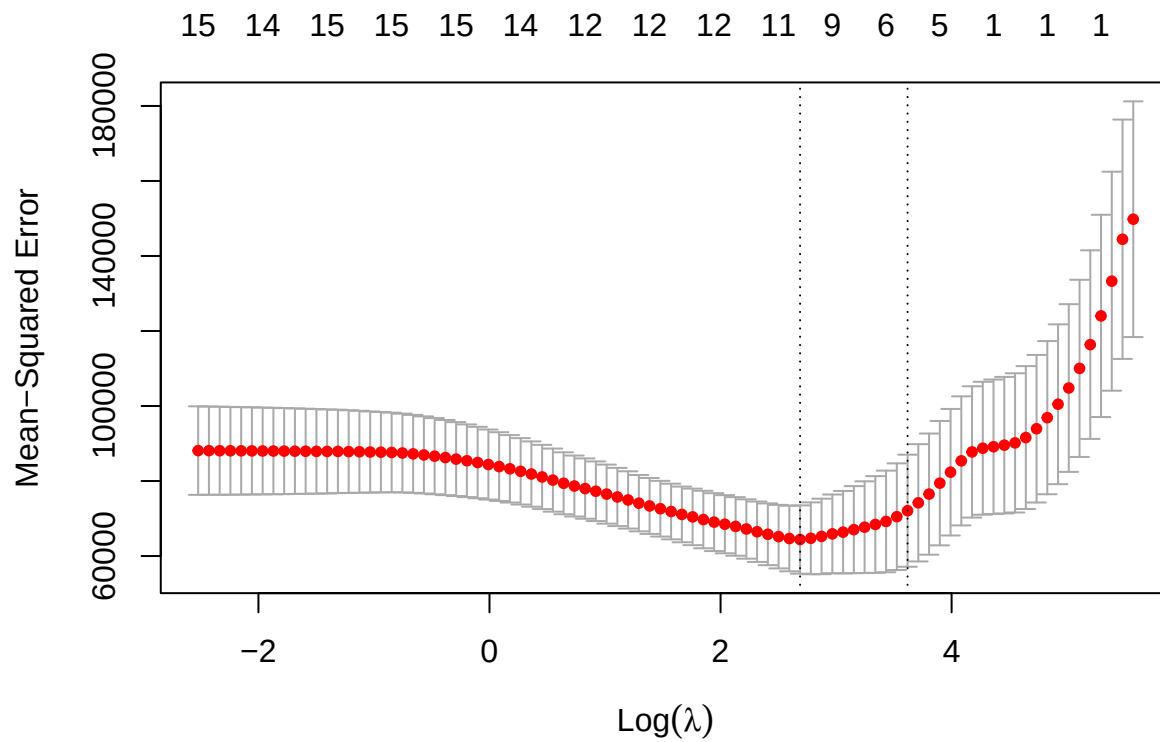
I will try using a different lasso model. For this model, I will change nlambda to 100. The nlambda parameter controls the number of lambda values when fitting the model. In other words, we try 100 different values of lambda, ranging from a large value, which focuses strong regularization, to a small value, which focuses on weak regularization.

```
model_lasso1 = cv.glmnet(x = as.matrix(uscrime[, -16]),
                        y = as.matrix(uscrime[, 16]),
                        alpha = 1,
                        nfolds = 8,
                        nlambda = 100,
                        type.measure = "mse",
                        family = "gaussian",
                        standardize = TRUE)
```

```
model_lasso1
```

```
##
## Call:  cv.glmnet(x = as.matrix(uscrime[, -16]), y = as.matrix(uscrime[,      16]), type.measure = "m
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min  14.71    32  64410  9102         10
## 1se  37.29    22  72082 14942          5
```

```
plot(model_lasso1)
```



```
model_lasso1$lambda.min
```

```
## [1] 14.70916
```

```
cbind(model_lasso1$lambda, model_lasso1$cvm, model_lasso1$nzzero)
```

```
##           [,1]      [,2] [,3]
## s0  263.09539664 149807.45    0
## s1  239.72272672 144461.05    1
## s2  218.42642038 133280.28    1
## s3  199.02201920 124026.01    1
## s4  181.34145155 116368.65    1
## s5  165.23157679 110034.78    1
## s6  150.55285890 104797.62    1
## s7  137.17815786 100469.08    1
## s8  124.99162839  96893.16    1
## s9  113.88771662  93940.51    1
## s10 103.77024576  91592.14    1
## s11  94.55158313  90168.53    1
## s12  86.15188108  89562.25    1
## s13  78.49838541  89145.83    1
## s14  71.52480520  88738.70    3
## s15  65.17073864  87756.76    4
## s16  59.38114984  85373.72    4
## s17  54.10589215  82352.19    5
## s18  49.29927381  79430.43    5
## s19  44.91966220  76515.42    5
## s20  40.92912321  74190.53    5
## s21  37.29309271  72082.23    5
```

|        |             |          |    |
|--------|-------------|----------|----|
| ## s22 | 33.98007714 | 70487.04 | 5  |
| ## s23 | 30.96138074 | 69225.28 | 6  |
| ## s24 | 28.21085701 | 68412.75 | 7  |
| ## s25 | 25.70468222 | 67696.91 | 9  |
| ## s26 | 23.42114910 | 67033.61 | 9  |
| ## s27 | 21.34047877 | 66362.14 | 9  |
| ## s28 | 19.44464945 | 65915.96 | 9  |
| ## s29 | 17.71724038 | 65231.61 | 9  |
| ## s30 | 16.14328958 | 64726.85 | 10 |
| ## s31 | 14.70916422 | 64409.60 | 10 |
| ## s32 | 13.40244262 | 64669.28 | 11 |
| ## s33 | 12.21180655 | 65176.78 | 11 |
| ## s34 | 11.12694332 | 65800.17 | 11 |
| ## s35 | 10.13845634 | 66480.53 | 11 |
| ## s36 | 9.23778382  | 67209.93 | 11 |
| ## s37 | 8.41712457  | 67898.61 | 11 |
| ## s38 | 7.66937042  | 68482.60 | 12 |
| ## s39 | 6.98804469  | 69044.84 | 12 |
| ## s40 | 6.36724606  | 69729.46 | 12 |
| ## s41 | 5.80159747  | 70439.32 | 12 |
| ## s42 | 5.28619954  | 71094.72 | 12 |
| ## s43 | 4.81658814  | 71871.42 | 12 |
| ## s44 | 4.38869572  | 72618.99 | 12 |
| ## s45 | 3.99881609  | 73358.84 | 12 |
| ## s46 | 3.64357229  | 74094.59 | 12 |
| ## s47 | 3.31988737  | 74952.94 | 12 |
| ## s48 | 3.02495773  | 75747.99 | 12 |
| ## s49 | 2.75622882  | 76510.42 | 12 |
| ## s50 | 2.51137305  | 77275.79 | 12 |
| ## s51 | 2.28826959  | 77991.55 | 12 |
| ## s52 | 2.08498602  | 78667.03 | 13 |
| ## s53 | 1.89976161  | 79398.69 | 13 |
| ## s54 | 1.73099203  | 80243.89 | 13 |
| ## s55 | 1.57721547  | 81083.56 | 13 |
| ## s56 | 1.43709999  | 81861.50 | 14 |
| ## s57 | 1.30943199  | 82585.56 | 14 |
| ## s58 | 1.19310566  | 83257.29 | 15 |
| ## s59 | 1.08711344  | 83848.52 | 15 |
| ## s60 | 0.99053728  | 84401.83 | 15 |
| ## s61 | 0.90254068  | 84929.30 | 15 |
| ## s62 | 0.82236145  | 85408.50 | 15 |
| ## s63 | 0.74930513  | 85843.08 | 15 |
| ## s64 | 0.68273892  | 86248.93 | 15 |
| ## s65 | 0.62208628  | 86623.10 | 15 |
| ## s66 | 0.56682185  | 86956.85 | 15 |
| ## s67 | 0.51646696  | 87236.92 | 15 |
| ## s68 | 0.47058546  | 87463.32 | 15 |
| ## s69 | 0.42877994  | 87606.00 | 15 |
| ## s70 | 0.39068831  | 87687.24 | 15 |
| ## s71 | 0.35598064  | 87733.39 | 15 |
| ## s72 | 0.32435630  | 87799.54 | 15 |
| ## s73 | 0.29554138  | 87847.16 | 15 |
| ## s74 | 0.26928630  | 87898.15 | 15 |
| ## s75 | 0.24536365  | 87886.61 | 15 |

```
## s76 0.22356622 87909.11 15
## s77 0.20370521 87938.91 15
## s78 0.18560860 87959.08 15
## s79 0.16911964 87975.25 15
## s80 0.15409552 87993.97 14
## s81 0.14040610 88012.69 14
## s82 0.12793281 88032.53 14
## s83 0.11656761 88046.78 14
## s84 0.10621207 88062.44 14
## s85 0.09677648 88077.84 14
## s86 0.08817912 88099.71 14
## s87 0.08034553 88116.54 15
```

```
coef(model_lasso1, s = model_lasso1$lambda.min)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept) -4149.1634647
## M           61.7709219
## So          45.3562792
## Ed          87.7668347
## Po1         104.7130086
## Po2          .
## LF          71.4832062
## M.F         15.9346818
## Pop          .
## NW          0.2580836
## U1           .
## U2          34.6955425
## Wealth      .
## Ineq        41.1423298
## Prob       -3388.8292437
## Time        .
```

```
min(model_lasso1$cvm)
```

```
## [1] 64409.6
```

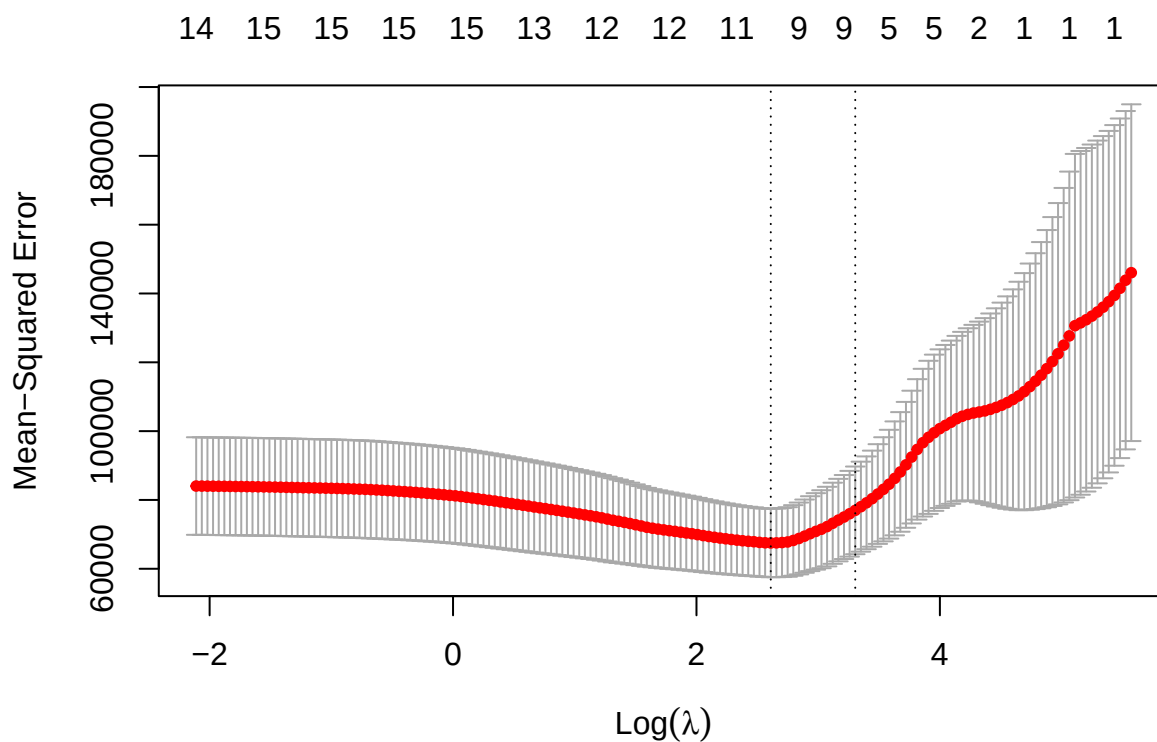
I created one last model where  $n_{\text{lambda}} = 200$ . Let's see what changes.

```
set.seed(123)
model_lasso2 = cv.glmnet(x = as.matrix(uscrime[, -16]),
                        y = as.matrix(uscrime[, 16]),
                        alpha = 1,
                        nfolds = 8,
                        nlambda = 200,
                        type.measure = "mse",
                        family = "gaussian",
                        standardize = TRUE)
```

```
model_lasso2
```

```
##  
## Call: cv.glmnet(x = as.matrix(uscrime[, -16]), y = as.matrix(uscrime[, 16]), type.measure = "m  
##  
## Measure: Mean-Squared Error  
##  
##      Lambda Index Measure      SE Nonzero  
## min  13.60    65  67491  9945      11  
## 1se  27.24    50  76973 12742       7
```

```
plot(model_lasso2)
```



```
model_lasso2$lambda.min
```

```
## [1] 13.60445
```

```
cbind(model_lasso2$lambda, model_lasso2$cvm, model_lasso2$nzzero)
```

```
##      [,1]      [,2] [,3]  
## s0  263.0953966 146048.61  0  
## s1  251.1960158 143860.05  1  
## s2  239.8348248 141512.74  1  
## s3  228.9874821 139442.77  1  
## s4  218.6307473 137622.48  1  
## s5  208.7424309 136026.77  1  
## s6  199.3013471 134632.91  1
```

|        |             |           |    |
|--------|-------------|-----------|----|
| ## s7  | 190.2872684 | 133420.30 | 1  |
| ## s8  | 181.6808819 | 132370.30 | 1  |
| ## s9  | 173.4637483 | 131466.02 | 1  |
| ## s10 | 165.6182625 | 130643.95 | 1  |
| ## s11 | 158.1276154 | 127683.98 | 1  |
| ## s12 | 150.9757582 | 124976.64 | 1  |
| ## s13 | 144.1473679 | 122500.00 | 1  |
| ## s14 | 137.6278147 | 120234.06 | 1  |
| ## s15 | 131.4031304 | 118160.56 | 1  |
| ## s16 | 125.4599786 | 116262.84 | 1  |
| ## s17 | 119.7856259 | 114525.71 | 1  |
| ## s18 | 114.3679151 | 112935.28 | 1  |
| ## s19 | 109.1952386 | 111480.86 | 1  |
| ## s20 | 104.2565139 | 110224.45 | 1  |
| ## s21 | 99.5411598  | 109207.22 | 1  |
| ## s22 | 95.0390735  | 108299.25 | 1  |
| ## s23 | 90.7406094  | 107540.95 | 1  |
| ## s24 | 86.6365578  | 106893.44 | 1  |
| ## s25 | 82.7181259  | 106343.53 | 1  |
| ## s26 | 78.9769184  | 105869.38 | 1  |
| ## s27 | 75.4049196  | 105565.72 | 2  |
| ## s28 | 71.9944766  | 105267.27 | 3  |
| ## s29 | 68.7382825  | 104855.60 | 3  |
| ## s30 | 65.6293607  | 104328.51 | 4  |
| ## s31 | 62.6610505  | 103611.69 | 4  |
| ## s32 | 59.8269922  | 102589.99 | 4  |
| ## s33 | 57.1211137  | 101662.65 | 5  |
| ## s34 | 54.5376178  | 100716.66 | 5  |
| ## s35 | 52.0709692  | 99530.54  | 5  |
| ## s36 | 49.7158832  | 98198.07  | 5  |
| ## s37 | 47.4673139  | 96645.72  | 5  |
| ## s38 | 45.3204438  | 94692.76  | 5  |
| ## s39 | 43.2706731  | 92461.53  | 5  |
| ## s40 | 41.3136103  | 90200.71  | 5  |
| ## s41 | 39.4450623  | 88142.92  | 5  |
| ## s42 | 37.6610257  | 86265.99  | 5  |
| ## s43 | 35.9576782  | 84578.83  | 5  |
| ## s44 | 34.3313703  | 83094.85  | 5  |
| ## s45 | 32.7786178  | 81719.89  | 5  |
| ## s46 | 31.2960937  | 80375.64  | 5  |
| ## s47 | 29.8806218  | 79168.99  | 6  |
| ## s48 | 28.5291695  | 78044.33  | 7  |
| ## s49 | 27.2388411  | 76973.13  | 7  |
| ## s50 | 26.0068722  | 75985.50  | 8  |
| ## s51 | 24.8306233  | 75034.10  | 9  |
| ## s52 | 23.7075742  | 74143.79  | 9  |
| ## s53 | 22.6353188  | 73219.98  | 9  |
| ## s54 | 21.6115598  | 72263.50  | 9  |
| ## s55 | 20.6341038  | 71496.00  | 9  |
| ## s56 | 19.7008565  | 70855.24  | 9  |
| ## s57 | 18.8098185  | 70200.00  | 9  |
| ## s58 | 17.9590808  | 69482.04  | 9  |
| ## s59 | 17.1468205  | 68804.75  | 9  |
| ## s60 | 16.3712974  | 68227.34  | 10 |

|         |            |          |    |
|---------|------------|----------|----|
| ## s61  | 15.6308500 | 67788.78 | 10 |
| ## s62  | 14.9238919 | 67618.90 | 10 |
| ## s63  | 14.2489083 | 67526.80 | 10 |
| ## s64  | 13.6044531 | 67491.38 | 11 |
| ## s65  | 12.9891456 | 67550.67 | 11 |
| ## s66  | 12.4016675 | 67663.34 | 11 |
| ## s67  | 11.8407601 | 67856.99 | 11 |
| ## s68  | 11.3052216 | 67984.21 | 11 |
| ## s69  | 10.7939046 | 68138.27 | 11 |
| ## s70  | 10.3057137 | 68316.98 | 11 |
| ## s71  | 9.8396029  | 68510.26 | 11 |
| ## s72  | 9.3945735  | 68708.75 | 11 |
| ## s73  | 8.9696721  | 68921.21 | 11 |
| ## s74  | 8.5639883  | 69149.04 | 11 |
| ## s75  | 8.1766529  | 69376.52 | 11 |
| ## s76  | 7.8068360  | 69642.64 | 11 |
| ## s77  | 7.4537454  | 69919.13 | 12 |
| ## s78  | 7.1166245  | 70186.26 | 12 |
| ## s79  | 6.7947511  | 70401.96 | 12 |
| ## s80  | 6.4874354  | 70632.96 | 12 |
| ## s81  | 6.1940192  | 70867.19 | 12 |
| ## s82  | 5.9138737  | 71071.06 | 12 |
| ## s83  | 5.6463987  | 71296.37 | 12 |
| ## s84  | 5.3910212  | 71528.60 | 12 |
| ## s85  | 5.1471940  | 71782.96 | 12 |
| ## s86  | 4.9143947  | 72062.11 | 12 |
| ## s87  | 4.6921245  | 72437.13 | 12 |
| ## s88  | 4.4799073  | 72799.22 | 12 |
| ## s89  | 4.2772883  | 73151.81 | 12 |
| ## s90  | 4.0838335  | 73485.38 | 12 |
| ## s91  | 3.8991283  | 73796.00 | 12 |
| ## s92  | 3.7227770  | 74092.14 | 12 |
| ## s93  | 3.5544018  | 74440.97 | 12 |
| ## s94  | 3.3936419  | 74805.12 | 12 |
| ## s95  | 3.2401529  | 75111.11 | 12 |
| ## s96  | 3.0936060  | 75396.91 | 12 |
| ## s97  | 2.9536872  | 75664.48 | 12 |
| ## s98  | 2.8200967  | 75932.74 | 12 |
| ## s99  | 2.6925483  | 76195.92 | 12 |
| ## s100 | 2.5707687  | 76463.22 | 12 |
| ## s101 | 2.4544969  | 76727.12 | 12 |
| ## s102 | 2.3434840  | 76981.84 | 12 |
| ## s103 | 2.2374920  | 77230.38 | 12 |
| ## s104 | 2.1362938  | 77477.59 | 13 |
| ## s105 | 2.0396727  | 77706.41 | 13 |
| ## s106 | 1.9474216  | 77952.56 | 13 |
| ## s107 | 1.8593429  | 78218.11 | 13 |
| ## s108 | 1.7752478  | 78462.16 | 13 |
| ## s109 | 1.6949562  | 78710.86 | 13 |
| ## s110 | 1.6182960  | 78948.31 | 13 |
| ## s111 | 1.5451031  | 79186.52 | 13 |
| ## s112 | 1.4752206  | 79428.56 | 13 |
| ## s113 | 1.4084987  | 79682.94 | 14 |
| ## s114 | 1.3447946  | 79906.25 | 14 |

|         |           |          |    |
|---------|-----------|----------|----|
| ## s115 | 1.2839717 | 80126.91 | 14 |
| ## s116 | 1.2258998 | 80347.39 | 15 |
| ## s117 | 1.1704543 | 80567.47 | 15 |
| ## s118 | 1.1175165 | 80785.38 | 15 |
| ## s119 | 1.0669731 | 81006.45 | 15 |
| ## s120 | 1.0187156 | 81196.28 | 15 |
| ## s121 | 0.9726407 | 81333.81 | 15 |
| ## s122 | 0.9286497 | 81498.47 | 15 |
| ## s123 | 0.8866484 | 81655.98 | 15 |
| ## s124 | 0.8465467 | 81797.81 | 15 |
| ## s125 | 0.8082588 | 81934.84 | 15 |
| ## s126 | 0.7717025 | 82060.87 | 15 |
| ## s127 | 0.7367997 | 82187.60 | 15 |
| ## s128 | 0.7034754 | 82303.33 | 15 |
| ## s129 | 0.6716584 | 82412.99 | 15 |
| ## s130 | 0.6412803 | 82516.79 | 15 |
| ## s131 | 0.6122763 | 82620.36 | 15 |
| ## s132 | 0.5845840 | 82724.05 | 15 |
| ## s133 | 0.5581442 | 82820.68 | 15 |
| ## s134 | 0.5329002 | 82911.83 | 15 |
| ## s135 | 0.5087980 | 83001.04 | 15 |
| ## s136 | 0.4857859 | 83077.89 | 15 |
| ## s137 | 0.4638146 | 83135.87 | 15 |
| ## s138 | 0.4428370 | 83192.14 | 15 |
| ## s139 | 0.4228082 | 83246.85 | 15 |
| ## s140 | 0.4036853 | 83298.00 | 15 |
| ## s141 | 0.3854272 | 83342.62 | 15 |
| ## s142 | 0.3679950 | 83382.81 | 15 |
| ## s143 | 0.3513512 | 83421.28 | 15 |
| ## s144 | 0.3354601 | 83458.92 | 15 |
| ## s145 | 0.3202878 | 83495.07 | 15 |
| ## s146 | 0.3058017 | 83530.08 | 15 |
| ## s147 | 0.2919708 | 83563.66 | 15 |
| ## s148 | 0.2787654 | 83596.44 | 15 |
| ## s149 | 0.2661573 | 83626.63 | 15 |
| ## s150 | 0.2541194 | 83663.33 | 15 |
| ## s151 | 0.2426260 | 83707.81 | 15 |
| ## s152 | 0.2316524 | 83741.28 | 15 |
| ## s153 | 0.2211752 | 83769.61 | 15 |
| ## s154 | 0.2111718 | 83796.10 | 15 |
| ## s155 | 0.2016208 | 83821.63 | 15 |
| ## s156 | 0.1925018 | 83845.82 | 15 |
| ## s157 | 0.1837953 | 83868.85 | 15 |
| ## s158 | 0.1754825 | 83891.14 | 15 |
| ## s159 | 0.1675457 | 83912.43 | 14 |
| ## s160 | 0.1599679 | 83933.09 | 14 |
| ## s161 | 0.1527328 | 83952.76 | 14 |
| ## s162 | 0.1458250 | 83971.37 | 14 |
| ## s163 | 0.1392295 | 83991.01 | 14 |
| ## s164 | 0.1329324 | 84009.05 | 14 |
| ## s165 | 0.1269201 | 84026.05 | 14 |
| ## s166 | 0.1211797 | 84043.30 | 14 |

```
coef(model_lasso2, s = model_lasso2$lambda.min)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept) -4283.2724257
## M           63.7512753
## So          49.4389539
## Ed          94.0054004
## Po1         105.0455899
## Po2         .
## LF          93.9238193
## M.F         15.8981244
## Pop         .
## NW          0.2285902
## U1         -141.9173491
## U2          40.6670522
## Wealth     .
## Ineq        42.3953795
## Prob       -3456.9135044
## Time       .
```

```
min(model_lasso2$cvm)
```

```
## [1] 67491.38
```

Based on all the models I created, the best model was the original model. The reason for this is that the RMSE is the lowest for that model. Since I am valuing predictive performance, a lower RMSE is ideal.

Lower RMSE means that the model has the smallest average prediction error.

## Elastic Net

```
set.seed(123)
model_enet <- cv.glmnet(
  x = x_scaled,
  y = y,
  alpha = 0.5,           # Elastic Net
  nlambda = 100,        # Grid of 100 lambda values
  nfolds = 10,          # 10-fold cross-validation
  type.measure = "mse", # Use MSE as performance metric
  family = "gaussian",
  standardize = TRUE    # Already scaled manually
)
```

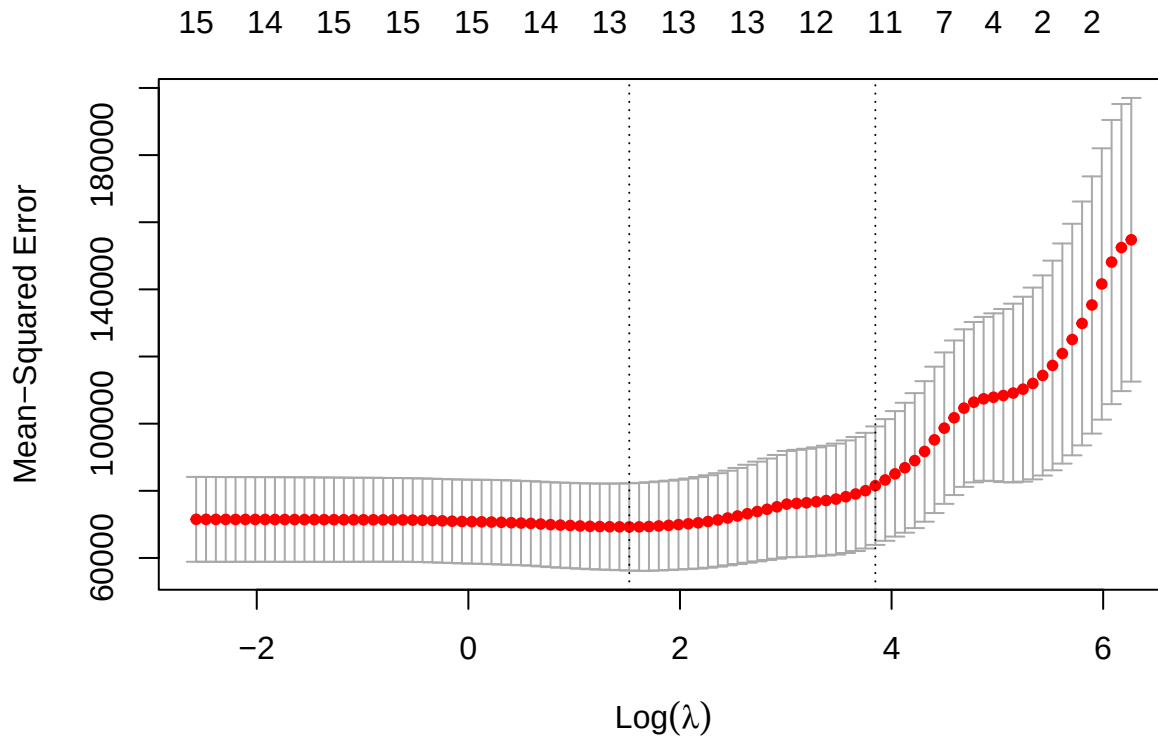
```
model_enet
```

```
##
```

```
## Call:  cv.glmnet(x = x_scaled, y = y, type.measure = "mse", nfolds = 10,      alpha = 0.5, nlambda =
##
```

```
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min   4.58    52  69248 12986      12
## 1se  46.84    27  81536 17645      11
```

```
plot(model_enet)
```



```
model_enet$lambda.min
```

```
## [1] 4.576539
```

```
cbind(model_enet$lambda, model_enet$cvm, model_enet$nzzero)
```

```
##           [,1]      [,2] [,3]
## s0  526.19079328 154776.27    0
## s1  479.44545344 152457.27    2
## s2  436.85284077 148138.80    2
## s3  398.04403841 141608.14    2
## s4  362.68290309 135353.85    2
## s5  330.46315358 129853.27    2
## s6  301.10571781 125058.51    2
## s7  274.35631572 120892.53    2
## s8  249.98325679 117338.06    2
## s9  227.77543324 114360.06    2
## s10 207.54049153 111970.35    2
## s11 189.10316626 110273.62    2
## s12 172.30376216 109120.06    3
```

|        |              |           |    |
|--------|--------------|-----------|----|
| ## s13 | 156.99677082 | 108377.67 | 3  |
| ## s14 | 143.04961040 | 107876.54 | 4  |
| ## s15 | 130.34147728 | 107402.06 | 4  |
| ## s16 | 118.76229967 | 106403.59 | 6  |
| ## s17 | 108.21178429 | 104640.18 | 6  |
| ## s18 | 98.59854762  | 101758.72 | 7  |
| ## s19 | 89.83932440  | 98668.65  | 7  |
| ## s20 | 81.85824643  | 95177.97  | 7  |
| ## s21 | 74.58618542  | 91728.28  | 7  |
| ## s22 | 67.96015428  | 88997.66  | 7  |
| ## s23 | 61.92276148  | 86873.49  | 7  |
| ## s24 | 56.42171402  | 85055.67  | 9  |
| ## s25 | 51.40936445  | 83233.54  | 11 |
| ## s26 | 46.84229820  | 81536.45  | 11 |
| ## s27 | 42.68095753  | 80043.85  | 11 |
| ## s28 | 38.88929890  | 79051.46  | 11 |
| ## s29 | 35.43448077  | 78245.27  | 11 |
| ## s30 | 32.28657916  | 77532.50  | 11 |
| ## s31 | 29.41832845  | 77097.93  | 11 |
| ## s32 | 26.80488523  | 76750.92  | 12 |
| ## s33 | 24.42361310  | 76450.42  | 12 |
| ## s34 | 22.25388663  | 76240.16  | 12 |
| ## s35 | 20.27691268  | 76036.98  | 13 |
| ## s36 | 18.47556764  | 75233.06  | 13 |
| ## s37 | 16.83424913  | 74499.22  | 13 |
| ## s38 | 15.33874084  | 73839.30  | 13 |
| ## s39 | 13.97608938  | 73177.13  | 13 |
| ## s40 | 12.73449211  | 72503.31  | 13 |
| ## s41 | 11.60319493  | 71900.44  | 14 |
| ## s42 | 10.57239907  | 71323.82  | 13 |
| ## s43 | 9.63317628   | 70861.55  | 13 |
| ## s44 | 8.77739145   | 70467.53  | 13 |
| ## s45 | 7.99763218   | 70175.08  | 13 |
| ## s46 | 7.28714458   | 69928.81  | 13 |
| ## s47 | 6.63977474   | 69711.37  | 12 |
| ## s48 | 6.04991545   | 69532.92  | 12 |
| ## s49 | 5.51245764   | 69365.94  | 12 |
| ## s50 | 5.02274610   | 69260.74  | 12 |
| ## s51 | 4.57653918   | 69247.85  | 12 |
| ## s52 | 4.16997205   | 69260.28  | 13 |
| ## s53 | 3.79952323   | 69304.07  | 13 |
| ## s54 | 3.46198405   | 69349.32  | 13 |
| ## s55 | 3.15443093   | 69423.19  | 13 |
| ## s56 | 2.87419998   | 69534.13  | 13 |
| ## s57 | 2.61886398   | 69666.13  | 13 |
| ## s58 | 2.38621132   | 69784.33  | 14 |
| ## s59 | 2.17422688   | 69923.43  | 14 |
| ## s60 | 1.98107456   | 70110.81  | 14 |
| ## s61 | 1.80508136   | 70284.99  | 15 |
| ## s62 | 1.64472290   | 70401.40  | 15 |
| ## s63 | 1.49861025   | 70501.01  | 15 |
| ## s64 | 1.36547785   | 70619.88  | 15 |
| ## s65 | 1.24417256   | 70698.19  | 15 |
| ## s66 | 1.13364370   | 70760.92  | 15 |

```
## s67 1.03293392 70807.86 15
## s68 0.94117091 70865.63 15
## s69 0.85755988 70957.17 15
## s70 0.78137663 71053.92 15
## s71 0.71196128 71153.09 15
## s72 0.64871260 71230.46 15
## s73 0.59108276 71290.64 15
## s74 0.53857260 71328.78 15
## s75 0.49072729 71345.61 15
## s76 0.44713243 71359.45 15
## s77 0.40741042 71372.58 15
## s78 0.37121720 71375.56 15
## s79 0.33823929 71377.95 15
## s80 0.30819104 71387.30 15
## s81 0.28081220 71401.80 15
## s82 0.25586562 71417.21 15
## s83 0.23313522 71428.99 15
## s84 0.21242413 71432.87 14
## s85 0.19355296 71441.73 14
## s86 0.17635825 71453.69 14
## s87 0.16069107 71463.17 14
## s88 0.14641571 71468.13 14
## s89 0.13340855 71469.77 14
## s90 0.12155690 71471.09 15
## s91 0.11075812 71473.10 15
## s92 0.10091867 71475.98 15
## s93 0.09195334 71486.52 15
## s94 0.08378445 71497.33 15
## s95 0.07634127 71506.53 15
```

```
coef(model_enet, s = model_enet$lambda.min)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept) 905.08511
## M           103.56327
## So           16.90215
## Ed           174.75228
## Po1          288.70149
## Po2           .
## LF           .
## M.F           55.49396
## Pop          -19.20869
## NW            19.02883
## U1           -75.76496
## U2           120.25967
## Wealth        58.91639
## Ineq          246.36956
## Prob         -91.13275
## Time           .
```

```
min(model_enet$cvm)
```

```
## [1] 69247.85
```

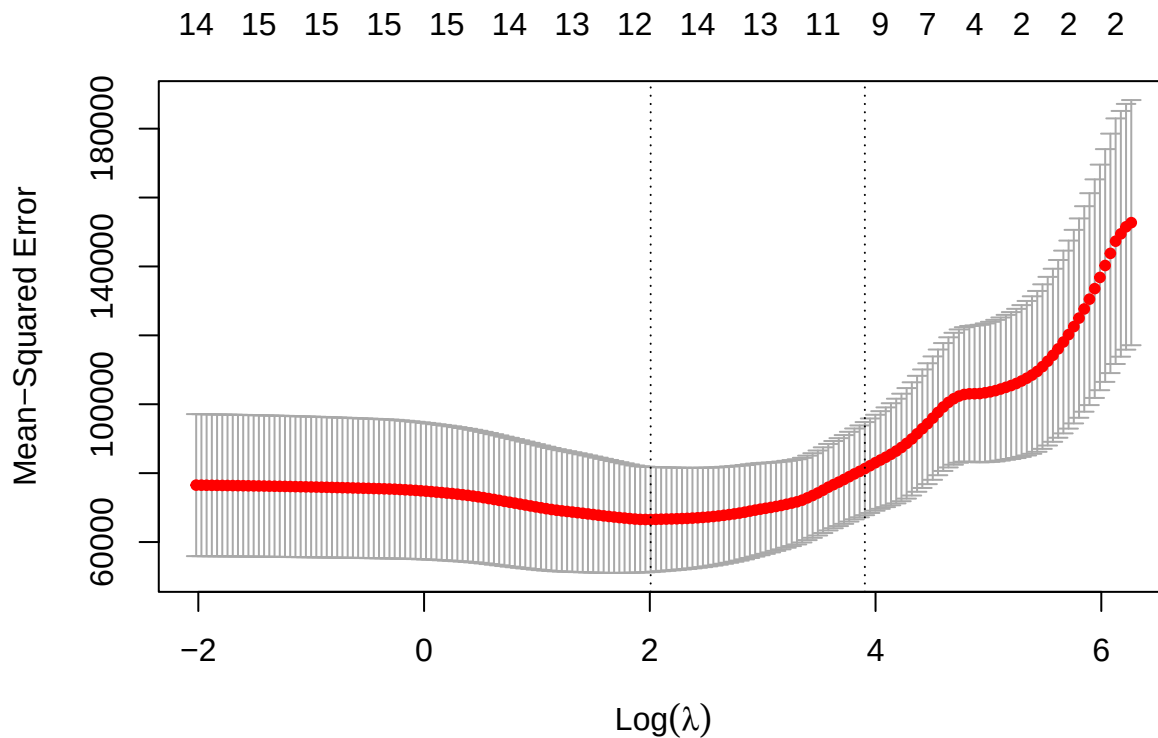
In this new model, I changed nlambd to 200 and nfolds to 15.

```
set.seed(123)
model_enet1 <- cv.glmnet(
  x = x_scaled,
  y = y,
  alpha = 0.5,           # Elastic Net
  nlambd = 200,          # Grid of 100 lambda values
  nfolds = 15,           # 10-fold cross-validation
  type.measure = "mse",  # Use MSE as performance metric
  family = "gaussian",
  standardize = TRUE     # Already scaled manually
)
```

```
model_enet1
```

```
##
## Call:  cv.glmnet(x = x_scaled, y = y, type.measure = "mse", nfolds = 15,      alpha = 0.5, nlambd = 
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min    7.45   93  66539 15248      13
## 1se   49.66   52  81312 13514      11
```

```
plot(model_enet1)
```



```
model_enet1$lambda.min
```

```
## [1] 7.445554
```

```
cbind(model_enet1$lambda, model_enet1$cvm, model_enet1$nzzero)
```

```
##           [,1]      [,2] [,3]
## s0  526.1907933 152718.15    0
## s1  502.3920316 151510.78    1
## s2  479.6696496 149470.28    2
## s3  457.9749643 147328.41    2
## s4  437.2614946 143797.12    2
## s5  417.4848618 140277.80    2
## s6  398.6026943 136815.22    2
## s7  380.5745367 133556.81    2
## s8  363.3617637 130509.03    2
## s9  346.9274967 127661.74    2
## s10 331.2365250 125004.85    2
## s11 316.2552308 122526.11    2
## s12 301.9515163 120217.75    2
## s13 288.2947358 118069.05    2
## s14 275.2556295 116071.54    2
## s15 262.8062609 114216.39    2
## s16 250.9199572 112495.08    2
## s17 239.5712519 110926.56    2
## s18 228.7358302 109539.81    2
## s19 218.3904772 108473.38    2
## s20 208.5130278 107557.92    2
## s21 199.0823196 106748.80    2
## s22 190.0781471 106033.95    2
## s23 181.4812188 105425.25    3
## s24 173.2731157 104910.16    3
## s25 165.4362519 104412.79    3
## s26 157.9538368 103949.18    3
## s27 150.8098393 103601.71    4
## s28 143.9889533 103331.72    4
## s29 137.4765650 103100.92    4
## s30 131.2587215 103052.14    4
## s31 125.3221010 103030.68    4
## s32 119.6539843 102835.54    6
## s33 114.2422274 102366.73    6
## s34 109.0752356 101589.40    6
## s35 104.1419384 100542.94    7
## s36  99.4317664  99198.25    7
## s37  94.9346278  97642.85    7
## s38  90.6408876  95973.92    7
## s39  86.5413463  94353.86    7
## s40  82.6272206  92899.94    7
## s41  78.8901246  91418.35    7
## s42  75.3220514  89938.87    7
## s43  71.9153564  88647.26    7
## s44  68.6627407  87453.33    7
## s45  65.5572356  86408.50    7
```

|        |            |          |    |
|--------|------------|----------|----|
| ## s46 | 62.5921875 | 85479.13 | 7  |
| ## s47 | 59.7612437 | 84613.12 | 8  |
| ## s48 | 57.0583389 | 83814.02 | 9  |
| ## s49 | 54.4776822 | 83016.02 | 10 |
| ## s50 | 52.0137444 | 82157.24 | 10 |
| ## s51 | 49.6612466 | 81311.58 | 11 |
| ## s52 | 47.4151484 | 80512.52 | 11 |
| ## s53 | 45.2706377 | 79777.27 | 11 |
| ## s54 | 43.2231197 | 78969.85 | 11 |
| ## s55 | 41.2682076 | 78158.43 | 11 |
| ## s56 | 39.4017131 | 77393.70 | 11 |
| ## s57 | 37.6196371 | 76667.09 | 11 |
| ## s58 | 35.9181615 | 75920.42 | 11 |
| ## s59 | 34.2936410 | 75053.11 | 11 |
| ## s60 | 32.7425948 | 74237.11 | 11 |
| ## s61 | 31.2617000 | 73503.58 | 11 |
| ## s62 | 29.8477837 | 72814.36 | 11 |
| ## s63 | 28.4978166 | 72159.15 | 12 |
| ## s64 | 27.2089062 | 71666.58 | 12 |
| ## s65 | 25.9782913 | 71309.03 | 12 |
| ## s66 | 24.8033350 | 70955.75 | 12 |
| ## s67 | 23.6815201 | 70616.69 | 12 |
| ## s68 | 22.6104431 | 70321.54 | 12 |
| ## s69 | 21.5878092 | 70041.16 | 12 |
| ## s70 | 20.6114274 | 69787.06 | 13 |
| ## s71 | 19.6792058 | 69538.89 | 13 |
| ## s72 | 18.7891470 | 69261.23 | 13 |
| ## s73 | 17.9393442 | 68974.47 | 13 |
| ## s74 | 17.1279765 | 68667.20 | 13 |
| ## s75 | 16.3533057 | 68402.15 | 13 |
| ## s76 | 15.6136721 | 68167.93 | 13 |
| ## s77 | 14.9074909 | 67958.64 | 13 |
| ## s78 | 14.2332491 | 67761.57 | 13 |
| ## s79 | 13.5895021 | 67575.64 | 13 |
| ## s80 | 12.9748709 | 67411.22 | 13 |
| ## s81 | 12.3880384 | 67268.43 | 13 |
| ## s82 | 11.8277473 | 67145.53 | 13 |
| ## s83 | 11.2927974 | 67043.08 | 14 |
| ## s84 | 10.7820424 | 66947.61 | 14 |
| ## s85 | 10.2943880 | 66862.18 | 13 |
| ## s86 | 9.8287894  | 66807.17 | 13 |
| ## s87 | 9.3842491  | 66753.94 | 13 |
| ## s88 | 8.9598146  | 66707.42 | 13 |
| ## s89 | 8.5545766  | 66667.45 | 13 |
| ## s90 | 8.1676669  | 66630.90 | 13 |
| ## s91 | 7.7982565  | 66581.26 | 13 |
| ## s92 | 7.4455539  | 66538.81 | 13 |
| ## s93 | 7.1088035  | 66548.22 | 13 |
| ## s94 | 6.7872838  | 66572.01 | 12 |
| ## s95 | 6.4803059  | 66667.12 | 12 |
| ## s96 | 6.1872121  | 66806.19 | 12 |
| ## s97 | 5.9073745  | 66951.76 | 12 |
| ## s98 | 5.6401934  | 67102.63 | 12 |
| ## s99 | 5.3850966  | 67261.73 | 12 |

|         |           |          |    |
|---------|-----------|----------|----|
| ## s100 | 5.1415373 | 67425.44 | 12 |
| ## s101 | 4.9089939 | 67599.01 | 12 |
| ## s102 | 4.6869680 | 67786.67 | 12 |
| ## s103 | 4.4749840 | 67974.50 | 12 |
| ## s104 | 4.2725877 | 68161.34 | 13 |
| ## s105 | 4.0793454 | 68348.39 | 13 |
| ## s106 | 3.8948432 | 68526.37 | 13 |
| ## s107 | 3.7186857 | 68702.04 | 13 |
| ## s108 | 3.5504956 | 68877.73 | 13 |
| ## s109 | 3.3899124 | 69046.38 | 13 |
| ## s110 | 3.2365921 | 69231.37 | 13 |
| ## s111 | 3.0902062 | 69450.29 | 13 |
| ## s112 | 2.9504412 | 69689.75 | 13 |
| ## s113 | 2.8169975 | 69941.08 | 13 |
| ## s114 | 2.6895892 | 70209.11 | 13 |
| ## s115 | 2.5679435 | 70478.88 | 13 |
| ## s116 | 2.4517995 | 70748.78 | 14 |
| ## s117 | 2.3409086 | 71016.91 | 14 |
| ## s118 | 2.2350330 | 71289.85 | 14 |
| ## s119 | 2.1339461 | 71556.10 | 14 |
| ## s120 | 2.0374312 | 71812.24 | 14 |
| ## s121 | 1.9452814 | 72096.63 | 14 |
| ## s122 | 1.8572995 | 72364.49 | 14 |
| ## s123 | 1.7732968 | 72625.31 | 15 |
| ## s124 | 1.6930935 | 72883.83 | 15 |
| ## s125 | 1.6165176 | 73125.35 | 15 |
| ## s126 | 1.5434051 | 73341.19 | 15 |
| ## s127 | 1.4735994 | 73544.92 | 15 |
| ## s128 | 1.4069508 | 73721.58 | 15 |
| ## s129 | 1.3433167 | 73898.78 | 15 |
| ## s130 | 1.2825607 | 74066.25 | 15 |
| ## s131 | 1.2245525 | 74208.81 | 15 |
| ## s132 | 1.1691680 | 74352.73 | 15 |
| ## s133 | 1.1162884 | 74491.09 | 15 |
| ## s134 | 1.0658005 | 74631.38 | 15 |
| ## s135 | 1.0175960 | 74758.03 | 15 |
| ## s136 | 0.9715718 | 74873.63 | 15 |
| ## s137 | 0.9276292 | 74980.96 | 15 |
| ## s138 | 0.8856740 | 75085.57 | 15 |
| ## s139 | 0.8456164 | 75171.96 | 15 |
| ## s140 | 0.8073705 | 75254.65 | 15 |
| ## s141 | 0.7708545 | 75336.55 | 15 |
| ## s142 | 0.7359900 | 75395.23 | 15 |
| ## s143 | 0.7027023 | 75443.82 | 15 |
| ## s144 | 0.6709202 | 75491.43 | 15 |
| ## s145 | 0.6405756 | 75536.03 | 15 |
| ## s146 | 0.6116034 | 75578.41 | 15 |
| ## s147 | 0.5839415 | 75619.03 | 15 |
| ## s148 | 0.5575308 | 75659.66 | 15 |
| ## s149 | 0.5323146 | 75699.92 | 15 |
| ## s150 | 0.5082389 | 75739.21 | 15 |
| ## s151 | 0.4852520 | 75777.08 | 15 |
| ## s152 | 0.4633049 | 75814.47 | 15 |
| ## s153 | 0.4423503 | 75851.10 | 15 |

```
## s154 0.4223435 75887.37 15
## s155 0.4032416 75922.65 15
## s156 0.3850037 75956.90 15
## s157 0.3675906 75989.72 15
## s158 0.3509650 76021.89 15
## s159 0.3350914 76053.57 15
## s160 0.3199358 76083.94 15
## s161 0.3054656 76113.21 15
## s162 0.2916499 76142.66 15
## s163 0.2784590 76170.83 15
## s164 0.2658648 76198.37 15
## s165 0.2538402 76224.93 15
## s166 0.2423594 76250.34 15
## s167 0.2313979 76275.15 15
## s168 0.2209321 76299.09 15
## s169 0.2109397 76321.87 14
## s170 0.2013992 76344.31 14
## s171 0.1922903 76366.25 14
## s172 0.1835933 76387.02 14
## s173 0.1752897 76407.43 14
## s174 0.1673616 76426.68 14
## s175 0.1597921 76445.34 14
## s176 0.1525650 76465.02 14
## s177 0.1456647 76487.36 14
## s178 0.1390765 76508.00 14
## s179 0.1327863 76524.65 14
```

```
coef(model_enet1, s = model_enet1$lambda.min)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept) 905.08511
## M           98.37010
## So          19.63621
## Ed          162.03243
## Po1         279.52175
## Po2          6.39211
## LF           .
## M.F         58.92884
## Pop        -11.02706
## NW          18.57127
## U1         -67.17689
## U2         107.90643
## Wealth      46.73384
## Ineq        225.30551
## Prob       -89.66047
## Time        .
```

```
min(model_enet1$cvm)
```

```
## [1] 66538.81
```